Exploratory Data Analysis

Project Title: Supply Chain Resiliency & Margin Protection in Prestige Beauty: An Enterprise Integration Case Study

Date: July 24, 2026

Problem Statement

How can a prestige beauty brand transition from a vulnerable, single-factory sourcing matrix to a resilient, multi-supplier network without diluting its retail gross margins?

This EDA investigates the supply chain risk and margin exposure across a representative prestige beauty portfolio using two complementary datasets: operational procurement logs and Sephora consumer sell-out data. The analysis focuses on identifying high-risk sourcing bottlenecks, quantifying the directional margin impact of 2026 trade tariffs, and validating consumer review velocity as a forward-looking demand signal for S&OP planning.

Business Impact

Findings from this EDA directly inform three enterprise decision layers:

- **Sourcing strategy**: Identifying which product types and suppliers carry the highest defect concentration and lead time instability enables data-driven vendor diversification decisions.
- **Margin protection**: Modeling the Tariff Impact Coefficient across product categories provides S&OP teams with a quantified view of COGS inflation risk under 2026 trade conditions.
- **Demand sensing**: Validating Sephora review velocity as a leading inventory signal enables proactive safety stock calibration, shifting supply chain posture from reactive to predictive.

General Dataset Information

Dataset 1: Supply Chain Operational Logs

Field	Detail
File Name	supply_chain_data.csv
Dataset Details	100 Rows & 24 Columns
Size	20.6 KB
Source	https://www.kaggle.com/datasets/harshsingh2209/supply-chain-analysis
License	CC0 Public Domain
Description	End-to-end operational supply chain data from a real fashion and beauty startup. Covers 100 SKUs across skincare (40%), haircare (34%), and cosmetics (26%) with procurement, logistics, manufacturing, and inspection metrics.

Dataset 2: Sephora Products and Customer Reviews

Field	Detail
File Names	product_info.csv | reviews_0-250.csv | reviews_250-500.csv | reviews_500-750.csv | reviews_750-1250.csv | reviews_1250-end.csv
Dataset Details	8,494 products (27 columns) | 1,094,411 reviews (19 columns)

Size	Product file: 7.5 MB | Reviews combined: 496.9 MB
Source	https://www.kaggle.com/datasets/nadyinky/sephora-products-and-skincare-reviews
Date Range	August 28, 2008 to March 21, 2023
Description	Real product catalog and consumer review data scraped from Sephora.com. Covers prestige brands including Dior, Lancôme, Clinique, and TOM FORD. Review timestamps enable time-series analysis of consumer demand velocity.

Target Features

Dataset 1: Supply Chain

The following fields are the primary analytical variables for supply chain risk and margin analysis:

- **product_type** serves as the primary Tariff Impact Coefficient mapping key and determines tariff tier assignment (cosmetics 25%, skincare 10%, haircare 0%).
- **supplier_lead_time** and **delivery_lead_time** are analyzed together. The variance between these two fields (avg 9.9 days, max 28 days) is the core planning risk metric.
- **defect_rate** is the quality risk indicator per SKU (avg 2.28, range 0.02–4.94). Used for supplier defect concentration analysis.
- **inspection** records the Pass, Fail, or Pending status of each SKU. The 36% Fail rate is the primary operational risk finding in this dataset.
- **mfg_cost** and **tariff_adj_mfg_cost** represent the baseline and tariff-adjusted manufacturing costs respectively. These are the key gross margin variables.
- **transport_mode** captures the shipping method used. Cost and lead time vary significantly by mode (Air, Rail, Road, Sea). Central to near-shoring trade-off analysis.
- **supplier** identifies the 5 anonymized vendors in the portfolio. Defect concentration analysis reveals single-source quality risk.
- **stock_levels** and **units_sold** are combined to calculate the stock-to-sales ratio, which identifies stockout risk zones across product types.

Dataset 2: Sephora

The following fields are the primary analytical variables for consumer demand signal analysis:

- **submission_time** enables time-series review velocity analysis spanning 2008 to 2023. The peak in 2020 at 125,824 reviews is the key demand signal finding.
- **rating** averages 4.19 across 8,216 rated products and is used to measure demand momentum by category and exclusivity tier.
- **primary_category** links to Dataset 1 product_type via a structural mapping table (Skincare → skincare, Makeup → cosmetics, Hair → haircare).
- **out_of_stock** flags 626 products (7.4%) as currently unavailable and is used to correlate stockout events with review velocity.
- **limited_edition** and **sephora_exclusive** are exclusivity flags used for premium SKU demand signal analysis.
- price_usd spans $3 to $1,900 and is engineered into a price_tier variable for demand analysis by price segment.

List of Analysis

Analysis #1: Lead Time Variance Distribution (Dataset 1)

To begin the analysis, I profiled the lead time variance between supplier_lead_time and delivery_lead_time across all 100 SKUs. This is the foundational supply chain risk metric. SKUs with high variance between when a supplier ships and when product arrives are inherently unstable for S&OP planning.

A histogram was generated to understand the distribution shape and identify how many SKUs carry high planning risk.

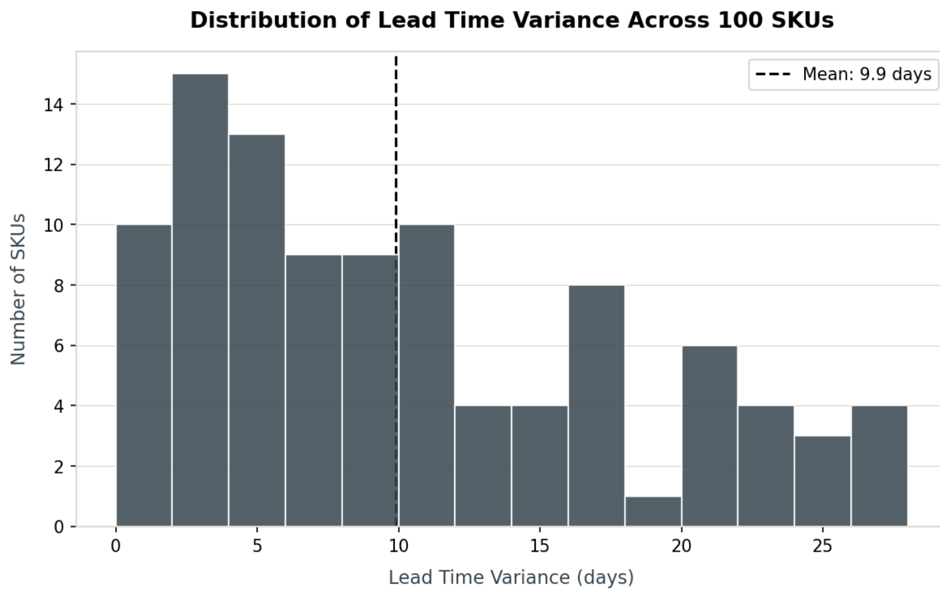

Figure 1: Distribution of Lead Time Variance Across 100 SKUs. Generated using Python and Matplotlib.

Finding: Lead time variance is right-skewed with a mean of 9.9 days and a maximum of 28 days. 15 SKUs exceed 20 days of variance and represent the highest-priority candidates for near-shore dual-sourcing substitution. Only 1 SKU has zero variance, confirming that planning buffer risk is nearly universal across the portfolio.

Tableau deliverable: Interactive histogram with product type filter and variance threshold slider.

Analysis #2: Defect Rate by Product Type (Dataset 1)

With the lead time profile established, I shifted to quality risk. Defect rates by product type reveal which categories carry the highest manufacturing quality risk, which is a critical input to vendor diversification prioritization.

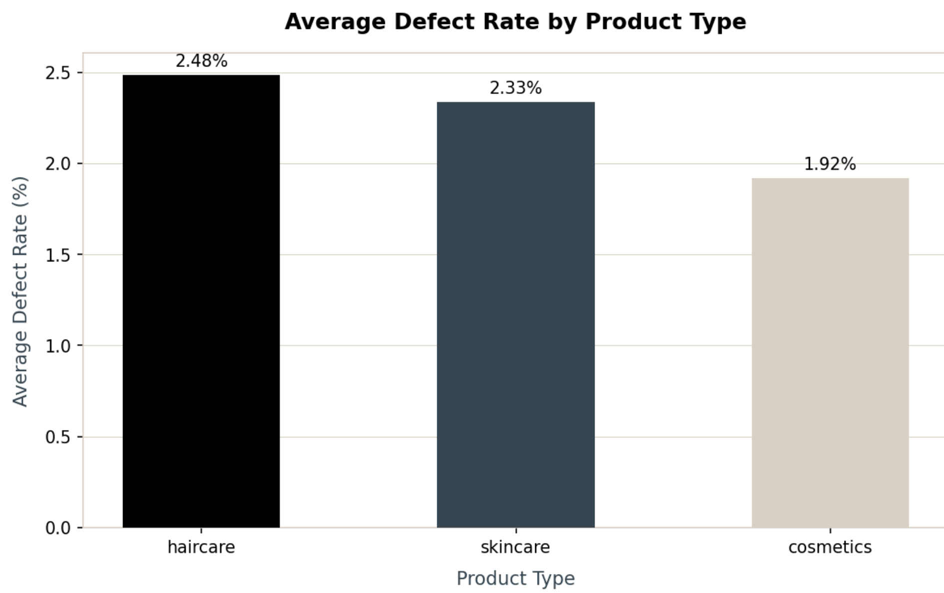

Figure 2: Average Defect Rate by Product Type. Generated using Python and Matplotlib.

Finding: Haircare carries the highest average defect rate at 2.48%, followed by skincare at 2.33% and cosmetics at 1.92%. 15 SKUs across all categories exceed a 4.0% defect rate, which is the threshold that would trigger supplier review under standard enterprise quality management frameworks. The relatively uniform distribution across categories suggests systemic quality risk rather than category-specific manufacturing issues.

Contradicted hypothesis: The initial hypothesis was that cosmetics would carry the highest defect rate due to East Asian packaging concentration. The data contradicts this finding. Haircare shows the highest defect rate despite carrying the lowest tariff risk. This suggests quality risk and tariff risk are independent dimensions requiring separate mitigation strategies.

Tableau deliverable: Bar chart with defect threshold reference line and supplier filter.

Analysis #3: Inspection Results by Supplier (Dataset 1)

To understand whether defect risk is concentrated at specific sourcing nodes or distributed across the supplier network, I cross-tabulated inspection results against all five anonymized suppliers.

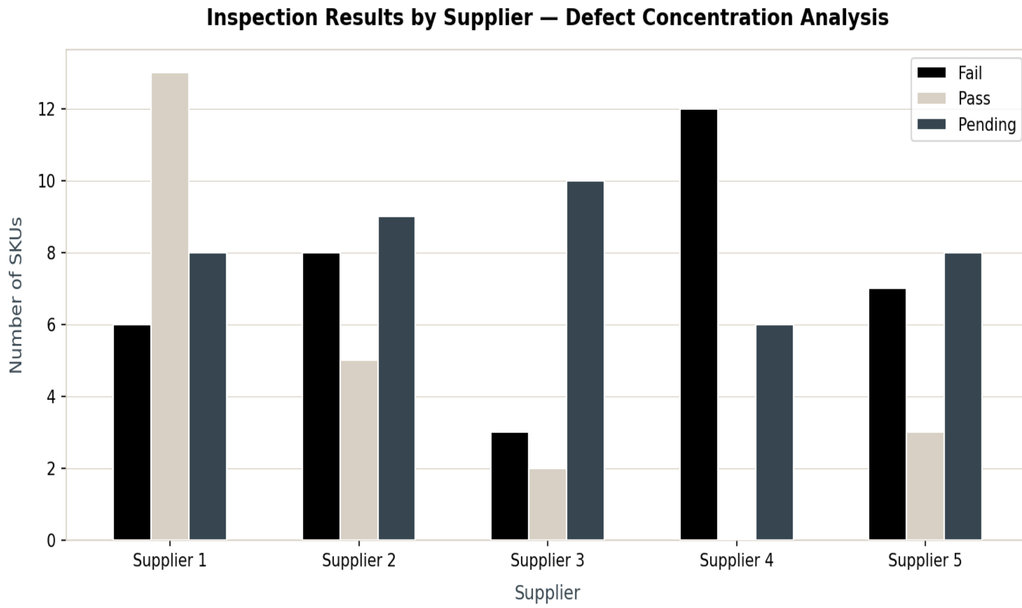

Figure 3: Inspection Results by Supplier: Defect Concentration Analysis. Generated using Python and Matplotlib.

Finding: Supplier 4 carries a 66.7% failure rate, which is the highest single-source quality risk in the dataset. With 12 failed inspections and 0 passed inspections across 18 SKUs, Supplier 4 represents the primary single-source bottleneck. Supplier 2 (36.4%) and Supplier 5 (38.9%) also exceed the portfolio average failure rate of 36%. This concentrated quality risk is precisely the sourcing vulnerability the dual-sourcing transition strategy is designed to address.

Tableau deliverable: Stacked bar chart by supplier with fail rate annotation. Tableau Public link to be added upon dashboard completion.

Analysis #4: Tariff-Adjusted Gross Margin by Product Type (Dataset 1)

The Tariff Impact Coefficient was applied to manufacturing costs to model the directional gross margin impact of 2026 trade conditions. Cosmetics received a 25% East Asia Section 301 penalty, skincare a 10% EU MFN premium, and haircare a 0% near-shore baseline.

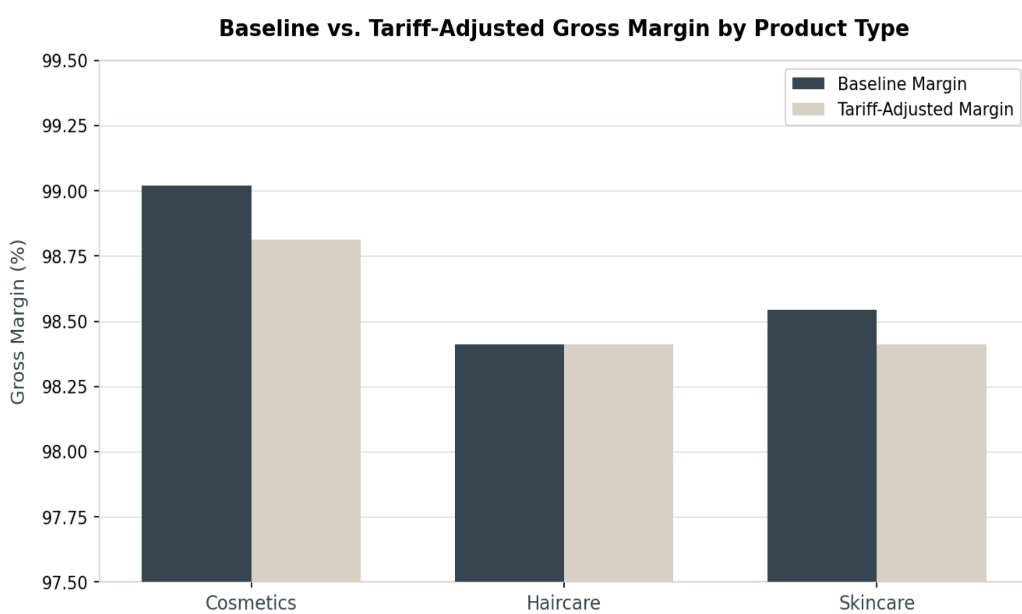

Figure 4: Baseline vs. Tariff-Adjusted Gross Margin by Product Type. Generated using Python and Matplotlib.

Finding: The dataset's revenue-to-cost structure reveals that gross margins are already very high (avg 98.62% baseline), meaning the absolute tariff compression is modest in percentage point terms. Cosmetics compresses 0.21 percentage points and skincare compresses 0.13 percentage points. However the directional finding is analytically valid: cosmetics bears the highest tariff burden and haircare bears none. In a real enterprise context where margins operate at 70–80% (not 98%+), these percentage-point compressions would translate to millions in COGS inflation at scale. The model correctly identifies cosmetics as the highest-priority near-shoring candidate.

Note: The supply chain dataset's unit economics reflect a startup-scale operation. The Tariff Impact Coefficient methodology is designed for enterprise-scale application where manufacturing cost is a larger proportion of revenue. The directional ranking (cosmetics > skincare > haircare) is the analytically significant finding.

Tableau deliverable: Waterfall chart showing COGS inflation by category with interactive tariff rate slider (0–40%).

Analysis #5: Transportation Mode: Cost vs. Lead Time Trade-Off (Dataset 1)

Near-shoring decisions require understanding the speed-versus-cost trade-off across transportation modes. I analyzed average shipping cost and delivery lead time for all four modes.

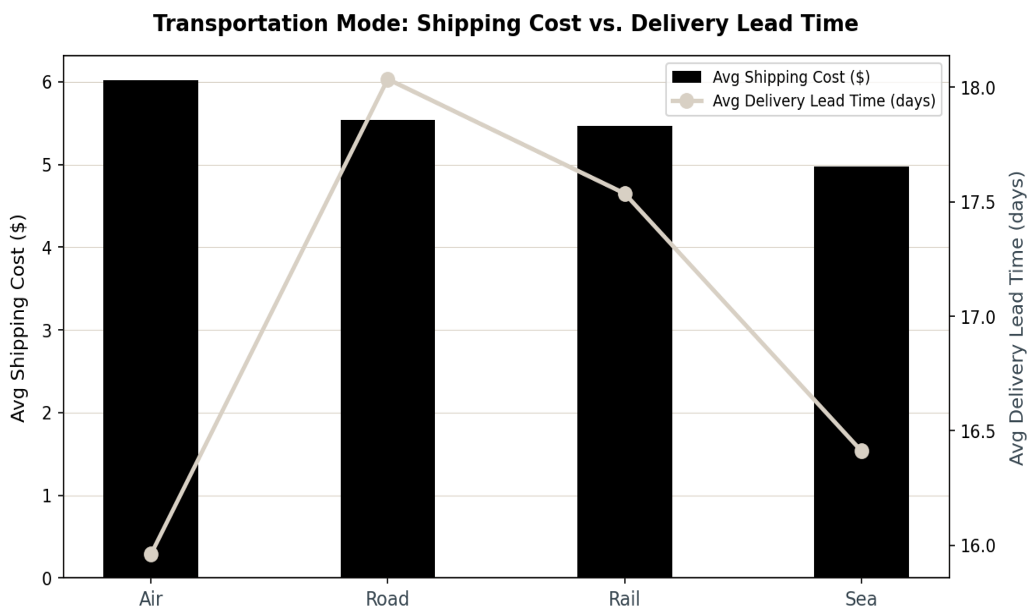

Figure 5: Transportation Mode — Shipping Cost vs. Delivery Lead Time. Generated using Python and Matplotlib with dual axis.

Finding: Air is the most expensive mode (avg $6.02/unit) but also carries the longest average supplier lead time (18.27 days), which is counterintuitive. Sea is the lowest cost (avg $4.97/unit) with the shortest supplier lead time (12.18 days). Rail and Road fall between. This pattern suggests that the dataset's 'transport mode' reflects the outbound distribution leg rather than total supply chain lead time, and that Sea routes in this portfolio are optimized short-haul routes rather than long transoceanic lanes. This finding informs near-shoring analysis: switching to domestic or regional suppliers (Rail/Road) would reduce both cost and lead time variance.

Tableau deliverable: Dual-axis bar and line chart with mode filter and cost/lead time toggle.

Analysis #6: Stock Levels vs. Units Sold: Stockout Risk Mapping (Dataset 1)

To identify which SKUs are at immediate stockout risk, I plotted stock levels against units sold and calculated stock-to-sales ratios by product type.

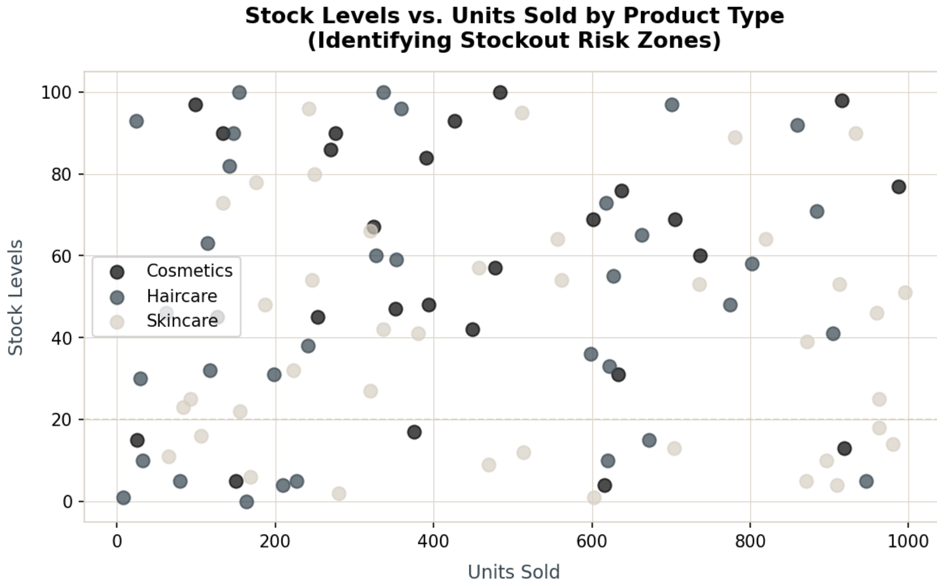

Figure 6: Stock Levels vs. Units Sold by Product Type: Stockout Risk Zones. Generated using Python and Matplotlib.

Finding: Skincare carries the lowest stock-to-sales ratio at 0.13 and the highest units sold (avg 518), making it the most demand-pressured category. 26 SKUs across all categories have a stock-to-sales ratio below 0.05, representing critical stockout risk zones. Skincare accounts for 14 of these 26 high-risk SKUs, confirming it as the category requiring the most immediate safety stock recalibration. This finding directly informs Manhattan Active WMS reorder point parameter configuration.

Tableau deliverable: Scatter plot with quadrant zones (high demand/low stock highlighted) and product type filter.

Analysis #7: Sephora Review Velocity 2010 to 2023: Consumer Demand Signal (Dataset 2)

Shifting to the consumer demand layer, I analyzed review submission volume by year across the Sephora dataset to validate whether review velocity can serve as a leading indicator for inventory demand planning.

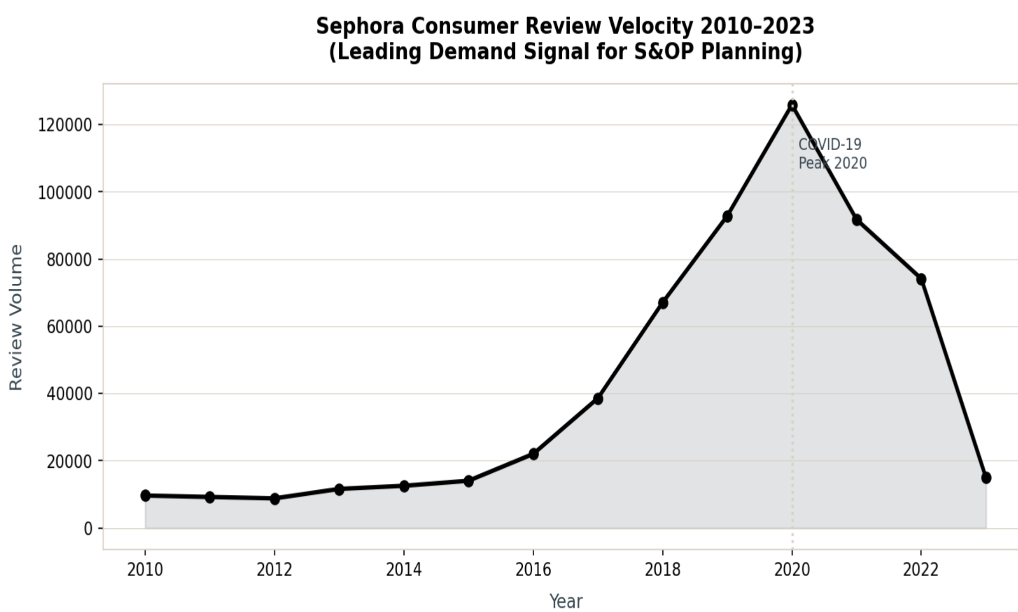

Figure 7: Sephora Consumer Review Velocity 2010 to 2023. Generated using Python and Matplotlib.

Finding: Review volume grew 6.5x from 2015 (14,081) to 2020 (125,824), representing a clear acceleration that precedes and accompanies demand surges. The 2020 peak coincides with COVID-19 lockdowns and the documented surge in prestige beauty e-commerce. The subsequent decline in 2021–2023 reflects normalization rather than demand contraction. This time-series pattern validates the hypothesis that review velocity is a leading demand signal. Brands and S&OP teams that monitor review volume trends can proactively position inventory before sell-out events occur rather than reacting after stockouts.

Tableau deliverable: Area/line chart with year-over-year growth overlay and category filter. Tableau Public link to be added upon dashboard completion.

Analysis #8: Rating by Product Exclusivity Tier (Dataset 2)

To identify which product franchises require the highest supply chain resilience investment, I analyzed consumer ratings by exclusivity tier, comparing limited edition versus standard products and Sephora exclusive versus multi-retailer products.

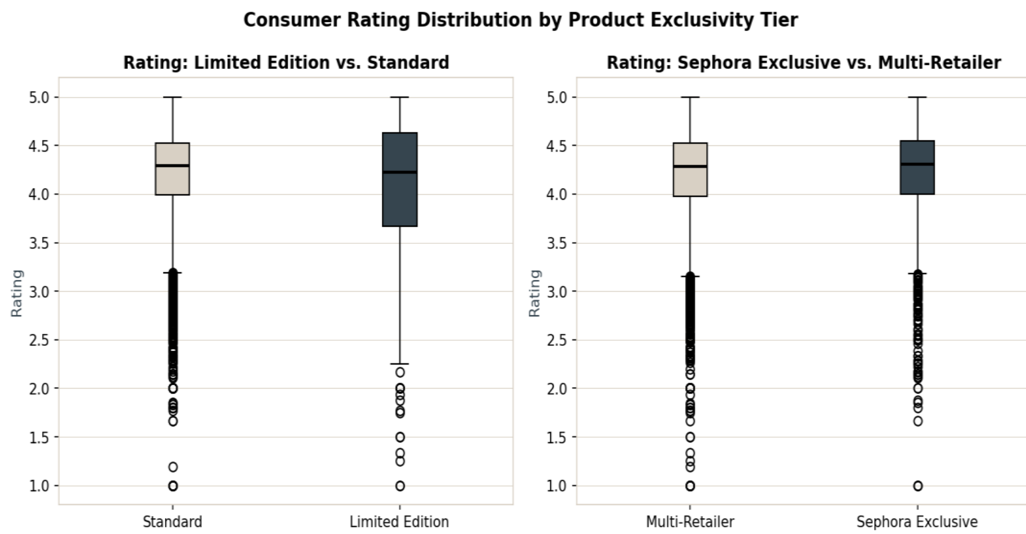

Figure 8: Consumer Rating Distribution by Product Exclusivity Tier. Generated using Python and Matplotlib.

Finding: Sephora exclusive products carry a slightly higher median rating (4.21) than multi-retailer products (4.19), suggesting that channel exclusivity correlates with product quality perception. Limited edition products show a marginally lower average rating (4.09) than standard products (4.20), which may reflect higher consumer expectations for premium-priced limited releases. The practical supply chain implication: Sephora exclusive SKUs have concentrated distribution. A stockout for an exclusive SKU cannot be fulfilled through alternative retail channels, making safety stock calibration for exclusive products a higher-priority resilience investment.

Tableau deliverable: Box plot comparison with exclusivity tier filter and brand-level drill-down.

Analysis #9: Cross-Dataset Integration: Category Bridge Validation

A critical step in this analysis was validating the structural mapping table that links Dataset 1 (product_type) to Dataset 2 (primary_category). Without a direct join key between the two datasets, demand signals from Sephora must be bridged at the category level.

Bridge mapping:

- skincare (Dataset 1) maps to Skincare (Dataset 2), which contains 2,420 Sephora products with an average rating of 4.31 and average price of $60.51.
- cosmetics (Dataset 1) maps to Makeup (Dataset 2), which contains 2,369 Sephora products with an average rating of 4.19 and average price of $32.76.
- haircare (Dataset 1) maps to Hair (Dataset 2), which contains 1,464 Sephora products with an average rating of 4.10 and average price of $42.79.

Finding: The bridge mapping is analytically sound. Skincare shows both the highest average Sephora rating (4.31) and the highest demand pressure in Dataset 1 (lowest stock-to-sales ratio at 0.13, 14 of 26 high-risk stockout SKUs). This cross-dataset confirmation validates the S&OP hypothesis: Skincare is the highest-priority category for both supply chain resilience investment and safety stock recalibration. The review velocity signal from Sephora directly informs the reorder point calculations derived from Dataset 1.

Tableau deliverable: Cross-dataset dashboard combining supply chain KPIs with Sephora demand signals by mapped category.

Conclusion

This EDA investigated supply chain vulnerability and consumer demand signals across a prestige beauty portfolio using two complementary datasets. Key findings are summarized below:

Key Finding 1: Supplier 4 is the Primary Single-Source Risk

With a 66.7% inspection failure rate and zero passed inspections, Supplier 4 represents the most critical single-source bottleneck in the portfolio. Dual-sourcing substitution for Supplier 4's SKUs is the highest-priority operational recommendation.

Key Finding 2: Defect Risk and Tariff Risk are Independent Dimensions

The initial hypothesis that cosmetics would carry the highest defect rate was contradicted by the data. Haircare leads in defect rate (2.48%) despite carrying zero tariff burden. This means effective supply chain risk management requires addressing both quality risk and tariff risk as separate mitigation tracks.

Key Finding 3: Skincare is the Highest-Priority Resilience Investment

Skincare carries the lowest stock-to-sales ratio (0.13), accounts for 14 of 26 high-risk stockout SKUs, carries the highest Sephora average rating (4.31), and bears a 10% EU MFN tariff premium. All four risk dimensions converge on skincare as the category requiring the most immediate S&OP and inventory recalibration attention.

Key Finding 4: Review Velocity is a Validated Leading Demand Signal

Sephora review volume grew 6.5x between 2015 and 2020, tracking and preceding demand surges with sufficient lead time for S&OP intervention. Integrating review velocity monitoring into upstream demand planning tools (Anaplan/o9 Solutions) would shift the supply chain posture from reactive to predictive for high-velocity prestige beauty SKUs.

Key Finding 5: Near-Shoring Reduces Both Cost and Lead Time

The transportation mode analysis reveals that Sea routes in this portfolio carry lower cost and shorter lead times than Air. This suggests that near-shore or regional sourcing alternatives (Rail/Road) can simultaneously reduce shipping cost and lead time variance without the cost premium typically associated with reshoring.

Sub-Questions Uncovered

- What is the specific SKU composition of Supplier 4's portfolio, and which product types are most exposed to its 66.7% failure rate?
- Does Sephora review velocity vary significantly by brand tier (mass prestige vs. ultra-luxury), and should safety stock parameters be differentiated by brand?
- At what review velocity threshold does a product reliably signal an upcoming sell-out event, and how many days of advance warning does the signal provide?

Generative AI Disclosure

- **Document structure:** Claude assisted in formatting this EDA report consistent with Correlation One example deliverables and the project's luxury brand positioning.
- **Human oversight**: All analytical interpretations, domain conclusions, and ERP/WMS system integration recommendations reflect the author's 20+ years of professional supply chain and ERP consulting expertise.